

Machine Learning (CE 40717)

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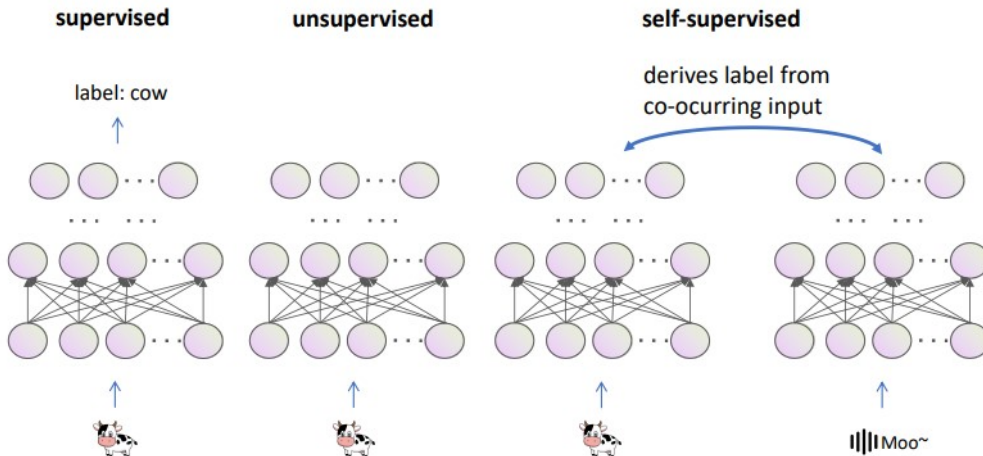
December 31, 2024



- ① Introduction
- ② Multimodality
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Setting



What is SSL

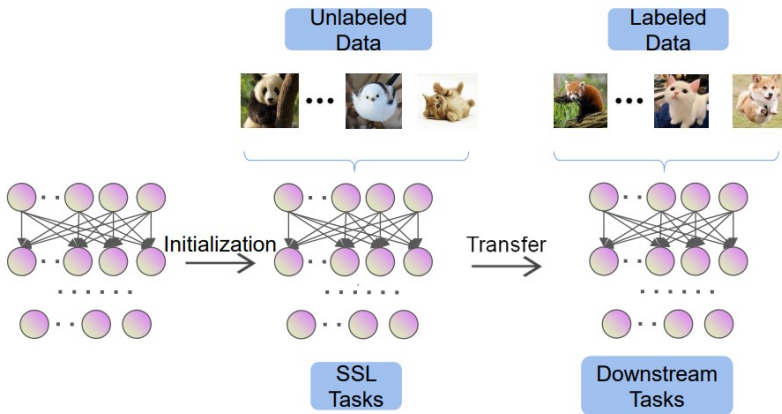
- Self-supervised learning defines a "pretext" task based on unlabeled inputs to produce descriptive and intelligible representations [Hastie et al., 2009, Goodfellow et al., 2016]
 - Labels of these pretext tasks are generated *automatically*
 - Can be used in other downstream tasks.

Labeled Data

Unlabeled Data

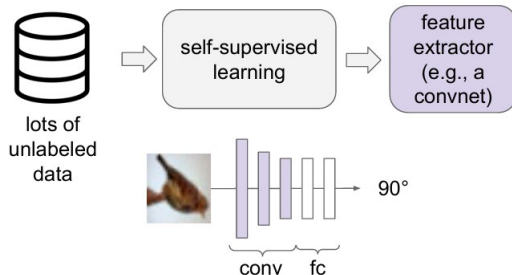
[reference]

General Pipeline

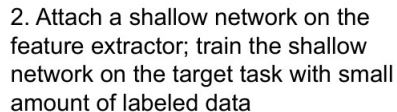


[reference]

Evaluation (Cont.)



1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations



Applications

- **NLP**

- Models like BERT, GPT, and T5 are based on SSL. They are pre-trained on massive text corpora using tasks like masked language modeling.
- Improving machine translation quality with SSL-based pre-training on multilingual corpora.

- **Computer Vision**

- Models like **SimCLR** and BYOL achieve state-of-the-art performance in image classification by learning from unlabeled images.
- Learning representations for tasks like disease diagnosis from limited annotated medical datasets.

Applications (Cont.)

- **Speech and Audio Processing**
 - Models like Wav2Vec and HuBERT use SSL in speech recognition task to learn representations from raw audio data. It's especially useful for languages with limited data.
 - Identifying individuals using voice features learned through SSL in speaker identification.
- **Robotics**
 - Leveraging SSL in reinforcement learning helps learn useful state representations for control tasks, enabling robots to perform complex actions.

Applications (Cont.)

- **Healthcare**

- Learning representations of DNA sequences to predict mutations or functional regions.
- Identifying diseases from clinical notes, imaging data, or time-series data such as ECGs.

- **Autonomous Vehicles**

- Integrating data from cameras, LiDAR, and radar through SSL.
- Using SSL for anomaly detection in vehicle systems.

Applications - Rotation (Cont.)^{1 2}

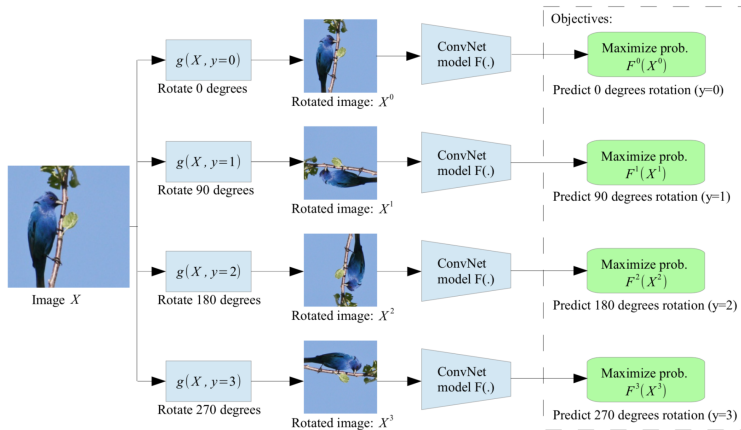


Figure 1: The model learns to predict which rotation is applied.

¹Some great examples

²Unsupervised Representation Learning by Predicting Image Rotations

3 4



Figure 2: The model learns to predict the relative position of two random patches.

³Some great examples

[illegible]

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5 6



index	permutation
64	9,4,6,8,3,2,5,1,7

Reorder patches according to the selected permutation

Figure 3: The model learns to solve jigsaw puzzle

⁵Some great examples

⁶Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles.

7 8



Figure 4: The model learns to track objects in videos.

⁷Some great examples

⁸Unsupervised Learning of Visual Representations using Videos

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Idea

- Don't learn from isolated images – take images together with some **context**
- **Video:** Image together with adjacent video frames
- **Sound:** Image with audio track from video
- **3D Image:** Image with depth map or point cloud
- **Language:** Image with natural-language text (e.g., captions or descriptions)

Why Language?

- **Rich Semantics**
 - Just a few words give rich information.
 - Acts as a bridge between sensory data and abstract human understanding.
- **Universality**
 - Language can describe almost any concept
 - Language can act as a **universal medium** for aligning other modalities, even structured data.

Why Language? (Cont.)

- **Large-Scale Data Availability**
 - The internet contains vast amounts of textual data.
 - Text data is relatively easier to collect, clean, and annotate (no need to experts) compared to modalities like video or audio.
 - Available datasets such as COCO (images and captions)
- **Pretrained Language Models (PLMs) as a Strong Foundation**
 - Large pretrained language models with remarkable capabilities.
 - Language models are highly transferable (transfer learning) across tasks, enabling multimodal systems to adapt to various downstream applications efficiently.

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Definition

- A machine learning technique for training models to distinguish between similar and dissimilar data points.
- **Key Idea**
 - Bring similar data points closer in the embedding space.
 - Push dissimilar data points farther apart.
- **Purpose:** Learn meaningful representations for downstream tasks like classification, clustering, or retrieval
- **Widely Used In:** Representation learning across domains such as computer vision, NLP, and multi-modal tasks.

Key Concepts

- **Embedding Space**
 - The data points are mapped into a high-dimensional space, called the embedding space.
 - Their relative positions encode similarity or dissimilarity.
- **Positive Pairs:** Data points that are semantically similar.
- **Negative Pairs:** Data points that are semantically different.
- **Objectives**
 - Minimize the distance between the embeddings of positive pairs.
 - Maximize the distance between the embeddings of negative pairs.

Common Components

- Dataset :
 - supervised: $D_m = \{(x_1^1, \dots, x_M^1, y^1), \dots, (x_1^n, \dots, x_M^n, y^n)\}$
 - self-supervised: $D_m = \{(x_1^1, \dots, x_M^1), \dots, (x_1^n, \dots, x_M^n)\}$
- The pseudo-label or signal generated for SSL can be denoted as $z = P(x_1, \cdot, x_M)$.
- Modality Encoder(s): $c = e_k(x_k^i; \theta_k)$ for each modality k .
- Fusion Module: f_ψ to integrate the encoded information of different modalities
- Pretext task head (like a predictive head) : g_γ and some SSL loss $\mathcal{L}_{SSL}$

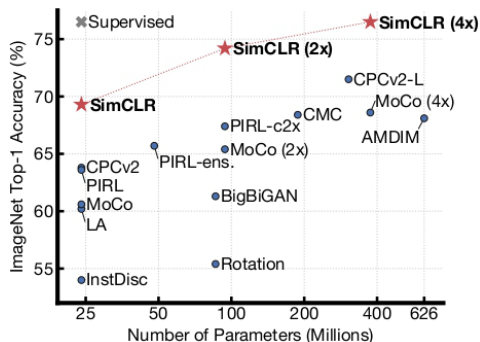
SimCLR^{9 10}

Figure 5: ImageNet Top-1 accuracy of linear classifiers trained on representations learned with different self-supervised methods (pretrained on ImageNet). Gray cross indicates supervised ResNet-50. SimCLR, is shown in bold.

⁹A Simple Framework for Contrastive Learning of Visual Representations

¹⁰Paper explained: A Simple Framework for Contrastive Learning of Visual Representations

SimCLR (Cont.)

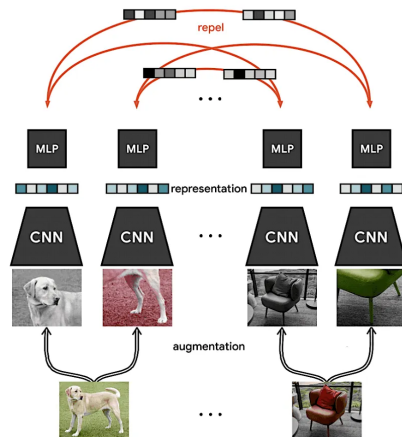


Figure 6: An illustration of the SimCLR training procedure.

SimCLR (Cont.)

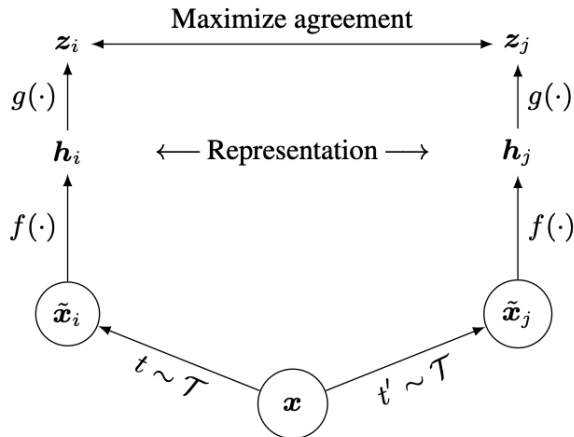
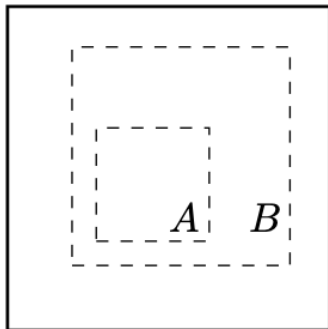
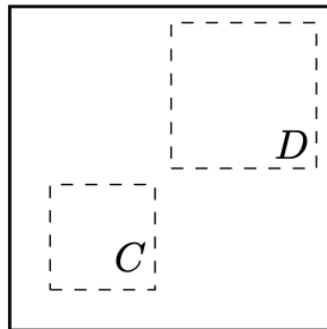


Figure 7: SimCLR overview

SimCLR (Cont.)



(a) Global and local views.



(b) Adjacent views.

Figure 8: Solid rectangles are images, dashed rectangles are random crops. By randomly cropping images, we sample contrastive prediction tasks that include global to local view ($B \rightarrow A$) or adjacent view ($D \rightarrow C$) prediction.

SimCLR (Cont.)



Algorithm 1 SimCLR's main learning algorithm.

input: batch size N , constant τ , structure of $f, g, \mathcal{T}$.
for sampled minibatch $\{\mathbf{x}_k\}_{k=1}^N$ **do**
 for all $k \in \{1, \dots, N\}$ **do**
 draw two augmentation functions $t \sim \mathcal{T}, t' \sim \mathcal{T}$
 # the first augmentation
 $\tilde{\mathbf{x}}_{2k-1} = t(\mathbf{x}_k)$
 $\mathbf{h}_{2k-1} = f(\tilde{\mathbf{x}}_{2k-1})$ # representation
 $\mathbf{z}_{2k-1} = g(\mathbf{h}_{2k-1})$ # projection
 # the second augmentation
 $\tilde{\mathbf{x}}_{2k} = t'(\mathbf{x}_k)$
 $\mathbf{h}_{2k} = f(\tilde{\mathbf{x}}_{2k})$ # representation
 $\mathbf{z}_{2k} = g(\mathbf{h}_{2k})$ # projection
 end for
 for all $i \in \{1, \dots, 2N\}$ and $j \in \{1, \dots, 2N\}$ **do**
 $s_{i,j} = \mathbf{z}_i^T \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$ # pairwise similarity
 end for
 define $\ell(i, j)$ as $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$
 $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$
 update networks f and g to minimize $\mathcal{L}$
end for
return encoder network $f(\cdot)$, and throw away $g(\cdot)$

Figure 9: Summarization of SimCLR method.

SimCLR (Cont.)

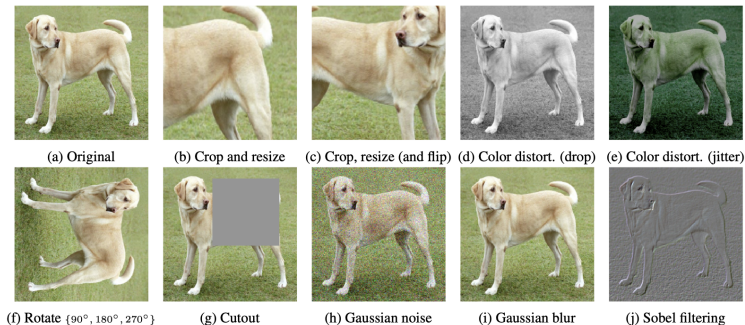


Figure 10: Illustrations of the studied data augmentation operators. Each augmentation can transform data stochastically with some internal parameters (e.g. rotation degree, noise level). Note that we only test these operators in ablation, the augmentation policy used to train our models only includes random crop (with flip and resize), color distortion, and Gaussian blur. (Original image cc-by: Von.grzanka)

SimCLR (Cont.)

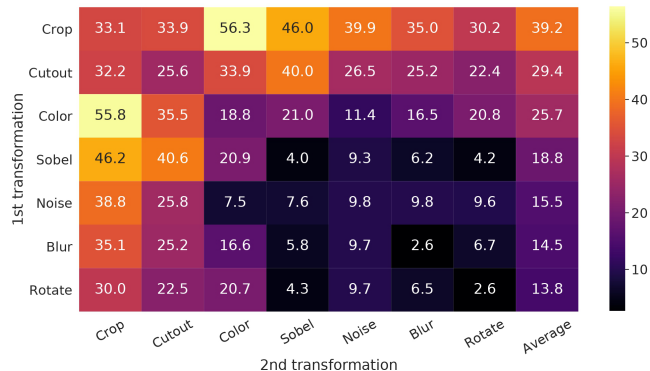


Figure 11: Linear evaluation (ImageNet top-1 accuracy) under individual or composition of data augmentations, applied only to one branch. For all columns but the last, diagonal entries correspond to single transformation, and off-diagonals correspond to composition of two transformations (applied sequentially). The last column reflects the average over the row.

CLIP

- Connecting text and images
- Contrastive Language–Image Pre-training
- CLIP $\Rightarrow$ a shared representation(embedding) between two modalities (text and images) by training on a large dataset of image-text pairs.

Architectures

- There many variations and structures

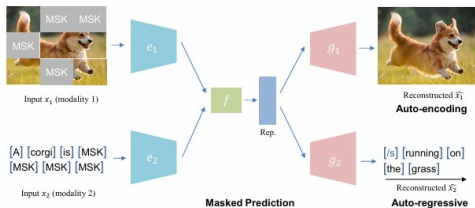


Figure 12: Figure 1 masked prediction frameworks

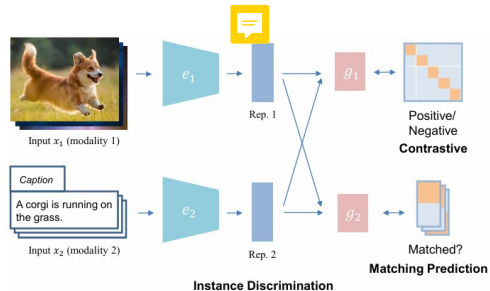
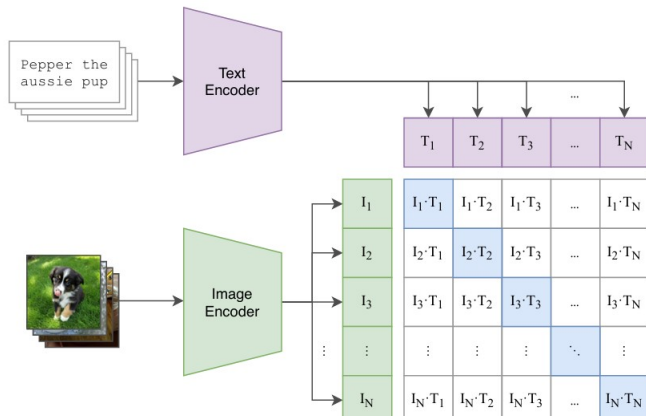


Figure 13: Figure 2 instance discrimination objectives

CLIP (Cont.)

- Image Encoder: a Vision Transformer (ViT) or a ResNet.
- Text Encoder: A Transformer model

(1) Contrastive pre-training



CLIP Goals

- CLIP was designed to mitigate a number of major problems:
- Costly datasets: Deep learning needs a lot of data, manually labeled datasets are expensive to construct.
 - CLIP learns from text–image pairs that are already publicly available on the internet
- Narrow: An ImageNet model excels at predicting the 1000 ImageNet categories but requires additional data and fine-tuning for other tasks.
 - CLIP can be adapted to perform a wide variety of visual classification tasks without needing additional training examples.

Loss Functions - Contrastive Loss^{11 12}

- Contrastive loss was first introduced in 2005 by Yann Le Cun et al.
- Its original application was in Dimensionality Reduction.

¹¹Dimensionality Reduction by Learning an Invariant Mapping

¹²Losses explained: Contrastive Loss

Loss Functions - Contrastive Loss (Cont.)

$$D_W(x^{(i)}, x''^{(i)}) = \|G_W(x^{(i)}) - G_W(x''^{(i)})\|_2$$

- $D_W(x^{(i)}, x''^{(i)})$ is dissimilarity between the two data points $x^{(i)}$ and $x''^{(i)}$.
- G_W is an embedding function parameterized by W .
- Generally, D_W can be any metric that indicates the dissimilarity between $x^{(i)}$ and $x''^{(i)}$.

Loss Functions - Contrastive Loss (Cont.)

$$L\left(W, y^i, x'^{(i)}, x''^{(i)}\right) = (1 - y^i) L_S\left(D_W(x'^{(i)}, x''^{(i)})\right) + y^i L_D\left(D_W(x'^{(i)}, x''^{(i)})\right)$$

- $(y^i, x'^{(i)}, x''^{(i)})$ is the i -th labeled sample pair.
- $Y = 0$ if $x'^{(i)}$ and $x''^{(i)}$ are deemed similar, and $Y = 1$ if they are deemed dissimilar.
- L_S is the partial loss function for a pair of similar points.
- L_D is the partial loss function for a pair of dissimilar points.
- L_S and L_D must be designed such that minimizing L with respect to W would result in low values of D_W for similar pairs and high values of D_W for dissimilar pairs.

Loss Functions - Contrastive Loss (Cont.)

$$\mathcal{L}(W) = \sum_{i=1}^P L\left(W, y^i, x'^{(i)}, x''^{(i)}\right)$$

- P is the number of training pairs (which may be as large as the square of the number of samples).

Loss Functions - InfoNCE Loss¹³

- First, we'll explore this loss from a theoretical perspective which has been discussed in its original paper.
- Next, we'll discuss how it can be applied in practice.
- It's the loss in its original paper:

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{\frac{p(x_{t+k}|c_t)}{p(x_{t+k})}}{\sum_{x_j \in X} \frac{p(x_t|c_t)}{p(x_t)}} \right]$$

¹³Representation Learning with Contrastive Predictive Coding

Loss Functions - InfoNCE Loss (Cont.)

- Let's start with mutual information.
- We have a set $X = \{x_1, \dots, x_N\}$ of N random samples containing one positive sample from $p(x_{t+k} | c_t)$ and $N - 1$ negative samples from the **proposal** distribution $p(x_{t+k})$
- Our purpose is to maximize mutual information:

$$I(x_{t+k}; c_t) = \sum_{x_{t+k}, c_t} p(x_{t+k}, c_t) \log \frac{p(x_{t+k} | c_t)}{p(x_{t+k})}$$

- c_t is context latent representation.

Loss Functions - InfoNCE Loss (Cont.)

- We know:

$$I(x_{t+k}; c_t) \leq \log N \rightarrow I(x_{t+k}; c_t) \geq \log N - \mathcal{L}_N$$

- $\mathcal{L}_N$ quantifies the gap between the true mutual information and the approximation.
- Minimizing $\mathcal{L}_N$ effectively maximizes the mutual information.

Loss Functions - InfoNCE Loss (Cont.)

- Categorical cross-entropy of classifying the positive sample correctly, with $\frac{f_k}{\sum_X f_k}$ being the prediction of the model.

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \right]$$

- We want to optimize it.

Loss Functions - InfoNCE Loss (Cont.)

- Let's write the optimal probability for this loss as $p(d = i | X, c_t)$ with $[d = i]$ being the indicator that sample x_i is the **positive** sample.
- The probability that sample x_i was drawn from the conditional distribution $p(x_{t+k} | c_t)$ rather than the proposal distribution $p(x_{t+k})$ can be derived as follows:

$$p(d = i | X, c_t) = \frac{p(x_i | c_t) \prod_{l \neq i} p(x_l)}{\sum_{j=1}^N p(x_j | c_t) \prod_{l \neq j} p(x_l)} = \frac{\frac{p(x_i | c_t)}{p(x_i)}}{\sum_{j=1}^N \frac{p(x_j | c_t)}{p(x_j)}}$$

- As we can see, the optimal value for $f_k(x_{t+k}, c_t)$ in $\mathcal{L}_N$ is proportional to $\frac{p(x_{t+k} | c_t)}{p(x_{t+k})}$ and this is independent of the choice of the number of negative samples $N - 1$.

Loss Functions - InfoNCE Loss (Cont.)

- We can evaluate the mutual information between the variables c_t and x_{t+k} as follows:

$$I(x_{t+k}, c_t) \geq \log(N) - \mathcal{L}_N$$

- It becomes tighter as N becomes larger.
- Minimizing the InfoNCE loss $\mathcal{L}_N$ maximizes a lower bound on mutual information.

Loss Functions - InfoNCE Loss (Cont.)

- In practice, we have:

$$\mathcal{L}_N = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(x_i, c_i) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(x_i, c_j) / \tau)}$$

- Used in models like SimCLR, MoCo, CLIP, and others.
- Next, we want to derive this formula from the theoretical one.

Loss Functions - InfoNCE Loss (Cont.)

- Step 1:**

$$\frac{p(x|c)}{p(x)} = \exp\left(\log\left(\frac{p(x|c)}{p(x)}\right)\right)$$

- But in practice, we rarely know the true densities $p(x|c)$ and $p(x)$.
- Instead, we learn a function that approximates their log-ratio.
- A common approach is to let a neural network produce embeddings $f(x)$ and $g(c)$.

Loss Functions - InfoNCE Loss (Cont.)

$$\log\left(\frac{p(x|c)}{p(x)}\right) \approx \text{sim}(f(x), g(c)) \xrightarrow{\text{we annotate it as}} \text{sim}(x, c) \rightarrow$$

$$\frac{p(x|c)}{p(x)} \approx \exp(\text{sim}(x, c)) \quad (1)$$

- $\text{sim}(x, c)$ is similarity function (e.g., cosine similarity or dot product).
- Replacing unknown densities with a similarity function, yielding a **softmax** function (which we'll discuss).
- It's straightforward to implement using standard deep-learning toolkits.

Loss Functions - InfoNCE Loss (Cont.)

- Why $\text{sim}(x, c)$ works?
 - It becomes large (positive) for the true **positive** pair (x, c) .
 - It becomes relatively small (negative) for **negative** pairs (x, c') .

$$\text{sim}(x, c) \gg \text{sim}(x, c') \longleftrightarrow p(x, c) \gg p(x, c')$$

- This is the property required to approximate the ratio $p(x | c) / p(x)$.

Loss Functions - InfoNCE Loss (Cont.)

- **Step 2:**
- In practice, we don't have the full distribution X or its expectations.
- Instead, we approximate this using batches of size N .
- Each x_{t+k} is treated as the **positive sample**, and the other x_j s in the batch are treated as **negative samples**.
- The expectation becomes a summation over batches:

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \right] \approx -\frac{1}{N} \sum_{i=1}^N \log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \quad (2)$$

Loss Functions - InfoNCE Loss (Cont.)

- **Step 3:**
- To control the sharpness of the similarity distribution, a temperature parameter τ is introduced:

$$\text{sim}(x, c) \rightarrow \frac{\text{sim}(x, c)}{\tau} \quad (3)$$

- τ helps balance gradients during training:
 - With no τ , large similarity scores might dominate the gradients, leading to unstable updates.
 - A carefully chosen τ scales the scores appropriately, ensuring stable convergence.

Loss Functions - InfoNCE Loss (Cont.)

- τ affects the distribution of similarity scores after applying the softmax function; in other words, it influences the sharpness of the softmax.
- Low τ :
 - High sharpness.
 - The softmax heavily favors the largest score.
 - The distribution becomes more concentrated on the top-scoring pair.
 - Encourages the model to focus strongly on the positive sample while ignoring negatives.
 - The loss becomes more sensitive to small differences in scores.
- High τ :
 - Low sharpness.
 - The softmax smooths the distribution, making it more uniform.
 - This encourages the model to consider a broader set of samples, not just the top-scoring pair.
 - Useful when the data is noisy or when the model needs to generalize better.

Loss Functions - InfoNCE Loss (Cont.)

- **Finally:** From equations (1) to (3), we derive:

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \right] \approx -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(x_i, c_i) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(x_i, c_j) / \tau)}$$

Applications

- CLIP enables searching for images by interpreting natural language descriptions.
- It classifies images into categories defined at inference using textual prompts (e.g., "This is a photo of a [label]").
- Visual Question Answering (VQA) becomes possible with CLIP's text-image alignment.
- Integrated into DALL·E or Stable Diffusion, it improves image generation with scoring and feedback loops.
- CLIP enhances robot-human interaction by understanding instructions about objects or scenes.

Applications

- It matches user queries with product images for seamless search and discovery in e-commerce.
- The model enables zero-shot classification to identify new or rare medical conditions.
- CLIP detects anomalies in manufacturing processes or equipment visuals to ensure quality.
- It personalizes digital experiences based on user preferences described in natural language.

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Contributions

These slides are authored by:

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